Obblivignes KanchanadeviVenkataraman

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Summary

Ph.D. student building physics-informed generative AI for spectroscopic single-molecule microscopy: diffusion models, attention U-Nets, Monte Carlo simulators and LLM-agent tools, judged by physical accuracy and transfer from simulation to experimental data.

Education

Ph.D., Computer Science — North Carolina State University — Aug 2025 – expected May 2029 · GPA 3.98/4.00
M.S., Computer Science — North Carolina State University — Aug 2023 – May 2025 · GPA 3.97/4.00
B.S., Computer Engineering — North Carolina State University — Aug 2020 – May 2023 · GPA 4.00/4.00 · Summa Cum Laude

Skills

Languages: Python, MATLAB, C, C++, Java, JavaScript, TypeScript, Bash
ML and data: PyTorch, Keras, scikit-learn, NumPy, SciPy, Pandas, Matplotlib
Methods: diffusion models (DDPM), attention U-Nets, LLM agents (GPT-4o, LiteLLM), Monte Carlo simulation, maximum-likelihood estimation
Tools: Git, SLURM, Docker, Kubernetes, React, Electron

Research Experience

Graduate Research Assistant — GIC Lab and MAP Lab, NC State University — Sep 2024 – present

- **DWP-sSMLM:** Building a dual-wedge-prism digital twin to recover molecular position and spectrum from one image.
- **FastSpecDiffuse:** Warm-start initialization from an SA_SpecUNet estimate cut diffusion sampling from 5–10 to 2–3 timesteps with no loss in denoising performance, outperforming naive DDPM respacing.
- **SpecDiffuse:** Built a lightweight conditional diffusion (DDPM) model on an SA_SpecUNet backbone in PyTorch; improved resolution ~50% on low-SNR data and transfers to experimental data.
- **SA_SpecUNet:** Added spatial attention gates, residual prediction and a multi-scale Perlin-noise background simulator, improving resolution ~25% over SpecUNet.
- **SpecUNet:** Developed grating-based Monte Carlo simulations (100k+ images) and a U-Net denoiser, improving resolution ~70% over conventional methods; *Analytical Chemistry* cover article.
- **SIMBA:** Led the denoising module of an LLM multi-agent imaging app, packaging the lab's denoisers as Python packages.

Graduate Research Assistant, Software Engineering AI — Dr. Bowen Xu's Lab, NC State — Jan 2024 – May 2024

- Trained a deep code-comment consistency checker on CodeXGLUE and shipped it in SEEDGuard with Docker images.

Undergraduate Research Assistant, Distributed Systems — Dr. Ruozhou Yu's Lab, NC State — Jan 2023 – Aug 2023

- Profiled microservices on a Kubernetes cluster; predicted latency with random forests ($R^2 = 0.96$).

Teaching Experience

Graduate Teaching Assistant, CSC 422: Automated Learning and Data Analysis — NC State — Jan 2026 – present

- Hold weekly office hours for ~60 students on ML concepts and a semester-long ML project; grade all coursework and design rubrics and answer keys.

Graduate Teaching Assistant, CSC 326: Software Engineering — NC State — Aug 2025 – Dec 2025

- Led two lab sections (~60 students) building a full-stack application; mentored teams on design, Git workflow and debugging.

Selected Publications

1. Mao, H., Mauny, H., **KanchanadeviVenkataraman, O.**, et al. (2026). SIMBA: An agentic AI platform for single-molecule multidimensional imaging. *bioRxiv*.
2. Mao, H., **KanchanadeviVenkataraman, O.**, et al. (2026). A Monte Carlo-based deep learning framework for accurate single-molecule spectroscopic imaging. *Proc. SPIE* 13858.
3. Mao, H., Liu, Y., **KanchanadeviVenkataraman, O.**, et al. (2025). Framework for accurate single-molecule spectroscopic imaging analyses using Monte-Carlo simulation and deep learning. *Analytical Chemistry*, 97(30). Cover article.

Awards and Service

Contributing researcher on $130K in NC State awards (Dean's Applied AI Research Accelerator, $100K; CMI Ideation, $30K) · Reviewer, RelKD 2026 Workshop at ACM SIGKDD 2026 · Poster, NC State 2026 Applied AI Symposium